

Subscription Performance Analysis & Retention Strategy



Introduction

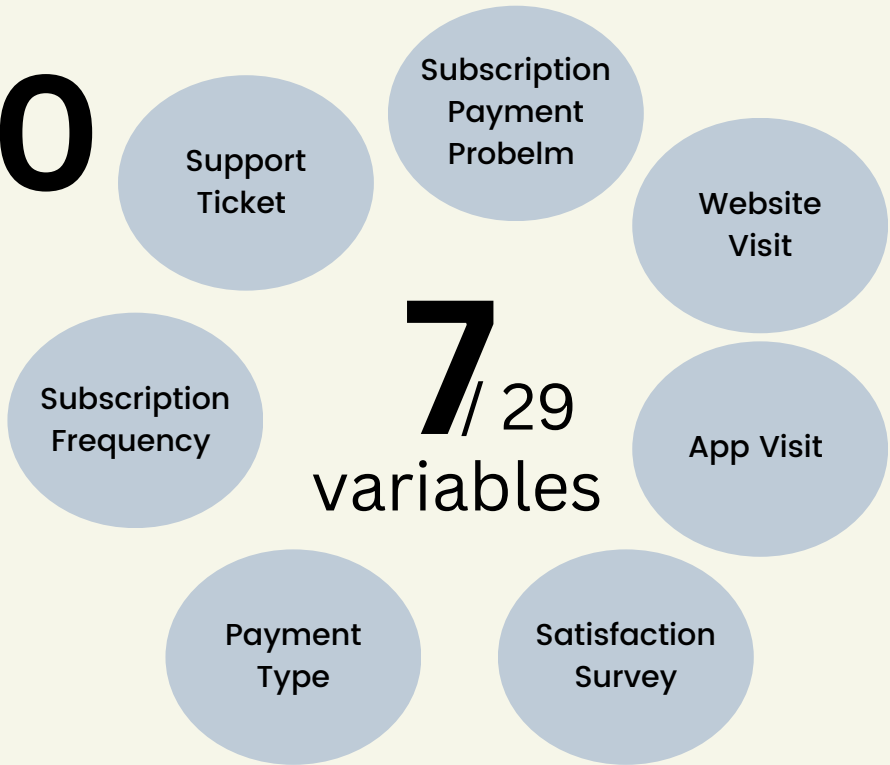
- Pet shop & dog Subscription Service
- Increasing churn rate
- Inadequate understanding of subscription performance

- Analyze subscription performance
- Identify churn reasons
- Recommend strategies to reduce churn



Data

200,000 customers



Methodology

Assumptions

Data assumption

- There is no missing values
- No major change in other quarter patterns

Model assumption

- Data is large enough for modelling
- Random forest ensures variables do not assume linear relationship

Consumer behaviour assumption

- Customers with bad customer support churn more
- Customers who faced payment problem are more likely to churn
- customers with lower interaction have higher frequency

Data Cleaning & Preparation



No missing value

Data Splitting

75% Training 25% Testing

Support ticket

Char

factor

Satisfaction Survey

Char

factor

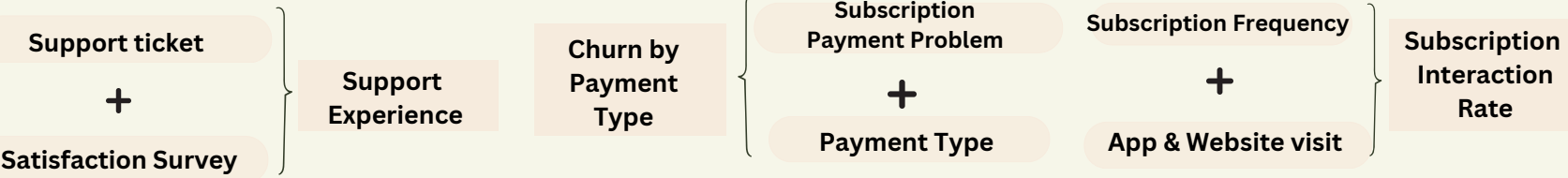
Payment Type

Char

factor

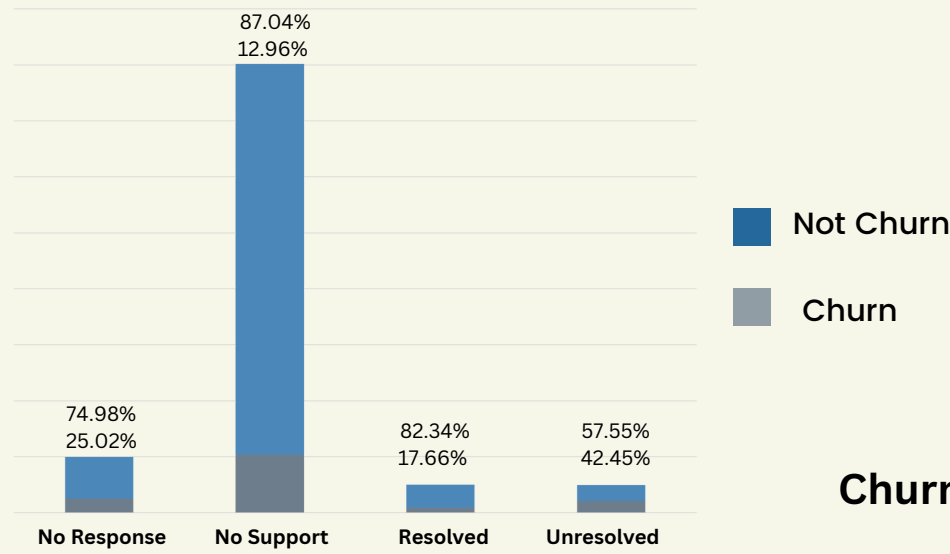
Techniques

Feature Engineering

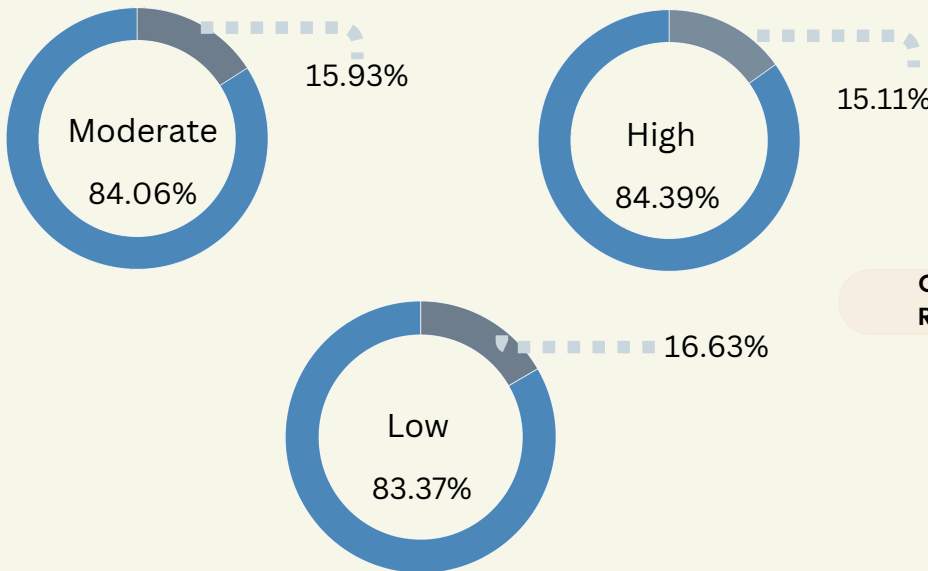


Support Experience

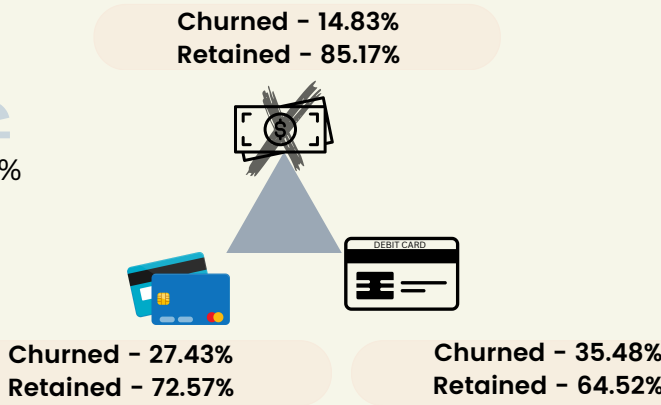
EDA



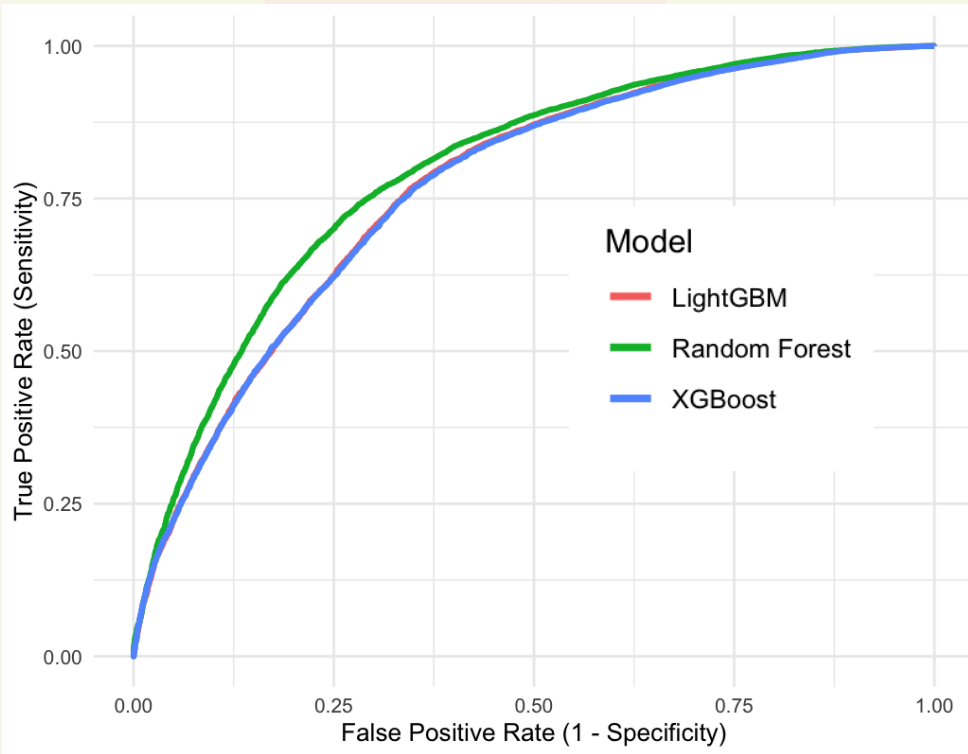
Subscription Interaction Rate



Churn by Payment Type



Predictive Analysis



	Accuracy	Sensitivity	PPV
Random Forest	0.757	0.684	0.360
XGBoost	0.812	0.495	0.421
LightGBM	0.812	0.502	0.422

Best model: Random Forest

AUC score: 0.794

CONFUSION MATRIX

True Positive
Churned
68.47%

True Negative
Churned but retained
77.06%

False Positive
Retained
22.94%

False Negative
Retained but Churned
31.53%

Key Findings

Subscription frequency affected app and website interaction.

Unresolved issues increase customer churn risk

Credit and debit card users churn significantly



Loyalty Points & Gift Cards



Flexible payment options



Survey Reward Incentive



Improve Customer Support

References

Belchior, L. M., António, N., & Fernandes, E. (2024). Online newspaper subscriptions: using machine learning to reduce and understand customer churn. Journal of Media Business Studies, 21(4), 364–387. <https://doi.org/10.1080/16522354.2024.2343638>